Cong Yin

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Last Updated: September 14, 2026

Bio

I am an early-career climate scientist pushing the boundaries of understanding wildfires and climate extremes using hydroclimatic, data-driven, and geostatistical approaches. My work has led to step changes in our understanding of the synchronicity and persistence of extreme fire weather, factors that strongly influence extreme fire activity. My recent research focuses on extreme wildfires, one of the most societally and environmentally destructive consequences of climate change. I concentrate on predicting extreme wildfires and developing a mechanistic understanding of their drivers, contributing to advances in fire science and fire management with the potential to save lives and protect property.

Education

Ph.D. in Geographic Information Science *Sep. 2018 – Jun. 2024*
University of Chinese Academy of Sciences, Beijing, China
Advisor: Dr. Juanle Wang
Dissertation: Detection, Changes, and Impacts of Global Compound Events

Visiting Ph.D. Student at Columbia University, New York, NY, USA *Apr. 2023 – Jun. 2024*
Department: Lamont-Doherty Earth Observatory
Advisor: Dr. Mingfang Ting, Dr. Kai Kornhuber

B.S. in Land Resources Management *Sep. 2014 – Jun. 2018*
Chang'an University, Xi'an, China

Appointments

Postdoctoral Scientist at University of California, Merced, CA, USA *Aug. 2024 – Present*
Department: Earth Institute
(formerly Sierra Nevada Research Institute)
Advisor: Dr. John Abatzoglou

Publications

Under Review / In Revision

[1] **C. Yin**, J. T. Abatzoglou, C. X. Cunningham, M. W. Jones, M. Sadegh, Energetically extreme fires drive disastrous societal impacts.

[2] **C. Yin**, M. Ting, K. Kornhuber, Drivers of the record-breaking concurrent heat extremes of 2023.

[3] J. T. Abatzoglou, K. C. Hegewisch, **C. Yin**, D. A. Kalashinkov, TerraClimate v1.1, An Updated Global Climate and Climatic Water Balance Dataset with Climate Scenarios.

[4] K. He, D. Zhao, X. Chen, J. Abatzoglou, C. Dong, A. AghaKouchak, C.-Y. Xu, Q. Liang, H. Chen, X. Tan, L. Brocca, J. Xia, **C. Yin**, V. P. Singh, G. MacDonald, Intensifying sub-seasonal wet-dry transitions amplify wildfire risk worldwide.

Published / Accepted

[5] **C. Yin**, J. T. Abatzoglou, P. Jain, M. Sadegh, M. K. Flannigan, M. W. Jones, Fire weather waves drive extreme fires globally. Commun. Earth & Environ. (2026)

[6] D. A. Kalashnikov, J. T. Abatzoglou, E. L. Williams, **C. Yin**, M. Gurazada, M. Kumar, A. P. Thomas, P. E. Ebiendele, Heatwaves enable wildfire activity in the western United States. Sci. Adv. (2026).

[7] **C. Yin**, J. T. Abatzoglou, M. W. Jones, A. C. Cullen, M. Sadegh, J. Wang, Y. Liu, Increasing synchronicity of global extreme fire weather. Sci. Adv. (2026).

[8] **C. Yin**, M. Ting, K. Kornhuber, R. M. Horton, Y. Yang, Y. Jiang, CETD, a global compound events detection and visualisation toolbox and dataset. Sci. Data 12, 356 (2025).

[9] Y. Liu, Y. Xin, **C. Yin**, A Transformer-based method to simulate multi-scale soil moisture. J. Hydrol. 655, 132900 (2025).

[10] **C. Yin**, Y. Yang, X. Chen, X. Yue, Y. Liu, Y. Xin, Global near real-time daily apparent temperature and heat wave dataset. Geosci. Data J. 10, 231–245 (2023).

[11] Y. Xin, Y. Yang, X. Chen, X. Yue, Y. Liu, **C. Yin**, One-kilometre monthly air temperature and precipitation product over the Mongolian Plateau for 1950–2020. Int. J. Climatol. 43, 3877–3891 (2023).

[12] **C. Yin**, Y. Yang, X. Chen, X. Yue, Y. Liu, Y. Xin, Changes in global heat waves and its socioeconomic exposure in a warmer future. Clim. Risk Manag. 38, 100459 (2022).

[13] Y. Xin, Y. Yang, X. Chen, X. Yue, Y. Liu, **C. Yin**, Evaluation of IMERG and ERA5 precipitation products over the Mongolian Plateau. Sci. Rep. 12, 21776 (2022).

[14] **C. Yin**, Y. Yang, F. Yang, X. Chen, Y. Xin, P. Luo, Diagnose the dominant climate factors and periods of spring phenology in Qinling Mountains, China. Ecol. Indic. 131, 108211 (2021).

[15] X. Chen, Y. Yang, **C. Yin**, Contribution of Changes in Snow Cover Extent to Shortwave Radiation Perturbations at the Top of the Atmosphere over the Northern Hemisphere during 2000–2019. Remote Sens. 13, 4938 (2021).

[16] **C. Yin**, F. Yang, J. Wang, Y. Ye, Spatiotemporal distribution and risk assessment of heat waves based on apparent temperature in the one belt and one road region. Remote Sens. 12, 1174 (2020).

[17] **C. Yin**, F. Yang, J. Wang, Analogs of Future Climate in Chinese Cities Identified in Present Observations. IEEE Access 8, 219151–219159 (2020).

Presentations

[1] Increasing Synchronicity of Global Extreme Fire Weather, AGU Fall Meeting — *Poster, Dec. 2025, New Orleans, USA*

[2] Fire Weather Waves Fuel Extreme Fires, AGU Fall Meeting — *Poster, Dec. 2025, New Orleans, USA*

[3] Strengthening Fire Preparedness and Coordination: Quantifying the Persistence and Synchronicity of Extreme Fire Weather — *Oral, Nov. 2025, Merced, USA*

[4] Increasing Global Intra-Regional and Inter-Regional Synchronous Fire Danger, 2025 AMS Denver Summit — *Poster, May 2025, Denver, USA*

[5] Increasing synchronicity of global extreme fire weather, SNRI Early Career Researcher Lightning Talks — *Oral, Mar. 2025, Merced, USA*

[6] CETD, a global compound events detection and visualization toolbox and dataset, AGU Fall Meeting — *Poster, Dec. 2023, San Francisco, USA*

Honors & Awards

[1] First Class Academic Scholarship, University of Chinese Academy of Sciences — *Oct. 2022 & 2023*

[2] First Class Director Scholarship, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences — *Oct. 2021 & 2022*

[3] Scholarship for the International PhD Joint Training Program, University of Chinese Academy of Sciences — *Nov. 2022*

[4] Merit Student Scholarship, University of Chinese Academy of Sciences — *May 2020, 2021 & 2023*

[5] Outstanding Undergraduate Thesis, Chang'an University — *Jun. 2018*

Reviewer, Media & Outreach

Review Activity

Earth's Future, Journal of Climate, Earth System Science Data, Communications Earth & Environment, International Journal of Wildland Fire, Fire, Journal of Geophysical Research: Machine Learning and Computation, Remote Sensing

Science Communication

Founder of the WeChat public account "Science of Extremes" — *2225 followers*

Media Interviews

[1] The New York Times, "Wildfire Seasons Are Starting to Overlap. That Spells Trouble for Firefighting." by Rebecca Dzombak — *Feb. 2026*

[2] The Associated Press, "Study finds warming world increases days when weather is prone to fires around the globe" by Seth Borenstein — *Feb. 2026*

[3] Vox, "Scientists have found another alarming pattern in wildfires" by Umair Irfan — *Feb. 2026*

[4] The Conversation, "Heat waves increase wildfire risk – a new study explains how much, and it's not a small number", Dmitri Kalashnikov, Cong Yin, Madhulika Gurazada, Mukesh Kumar — *Jun. 2026*

[5] The Debrief, "'Fire Weather Synchronicity' is Increasing Globally, as Climate Change Strains International Fire Fighting Resources" by Ryan Whalen — *Feb. 2026*

Selected Coverage

The New York Times, The Washington Post, The Guardian, The Associated Press, U.S. News & World Report, ABC News, Nature Briefing, University of California